Research Question: Among used vehicles listed on Craigslist in the U.S. in April-May 2021, is the decline in listing price associated with an additional 10,000 miles smaller for pickup trucks than for sedans, after accounting for the vehicle age, manufacturer, condition, transmission and drivetrain? We will measure and compare the change in listing price associated with each additional 10000 miles for the two vehicle types?

Hypothesis: We expect listing prices to decrease as mileage increases for both pickup trucks and sedans, but we expect this decline to be smaller for pickup trucks than sedans. More specifically, we expect the price decrease associated with 10000 miles should be less negative for pickups than for sedans after accounting for the other vehicle characteristics. Evidence would support our hypothesis if pickups show a consistently smaller decline per 10000 miles; similar declines or larger declines would challenge it. We expect this because pickups are designed for heavier and long term use which makes buyers less sensitive to an increase in mileage.

Logical Method

We will use a data set from kaggle linked here. It is a currently accessible dataset containing U.S. craigslist vehicle listings. We will use records classified as either pickup or sedan and the fields for price, odometer mileage, model year, manufacturer, condition, transmission, and drivetrain. One object in our analysis indicates one individual Craigslist vehicle listing.

From the raw listings, we will first extract pickup and sedan records and the fields needed for our question. We will then identify missing, duplicate and implausible observations and produce a cleaned set of listings with usable price, mileage, and vehicle-characteristic information. From the cleaned data, we will compute vehicle age from model year and express odometer mileage in units of 10000 miles, producing a consistent set of quantities for comparing vehicles. Using these quantities, we will compare how listing price changes as mileage increases while accounting for differences in vehicle age, manufacturer, condition, transmission, and drivetrain. This produces our final quantity of interest: the adjusted price decline associated with an additional 10000 miles for each vehicle type and the difference between those declines.
Consistently smaller declines for pickups vs sedans is what we would be looking for in order to support our hypothesis, similar or greater declines would challenge this.

By Checkpoint 1, we will have loaded the dataset, extracted and cleaned the required records, documented missing and implausible values, and created vehicle-age and 10,000-mile mileage measures. Our artifacts will include the cleaned dataset, a summary of preprocessing decisions, and exploratory summaries of the relevant variables.

By Checkpoint 2, we will represent each vehicle using continuous numerical features and encoded categorical features and use multiple linear regression with an interaction between mileage and vehicle type. The interaction term will measure whether the price change associated with an additional 10,000 miles differs between pickups and sedans. We will examine both the magnitude and statistical significance of this interaction; a positive, statistically significant interaction would support our hypothesis that pickup prices decline less rapidly with mileage. Our primary visualization will be a grouped scatter plot of mileage against listing price with separate trend lines for pickups and sedans. Visibly different slopes would illustrate a difference in the mileage-price relationship, while approximately parallel trend lines would suggest similar mileage effects. The Checkpoint 2 artifacts will therefore include the estimated price decline for each vehicle type, the interaction result, and the grouped scatter plot with trend lines.